A Machine Learning Framework for Predicting Temperature Anomalies and Global Warming Trends in Nepal using Earth Observation Data

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I. Introduction

The rapid accumulation of greenhouse gases in the Earth's atmosphere has undeniably accelerated the pace of global warming, bringing about significant alterations to the global climate system. Among the most profoundly impacted regions is the Himalayas of Nepal, which, despite contributing a negligible fraction to the global carbon footprint, is bearing a disproportionate brunt of climate change [1]. Rising temperatures and erratic weather patterns have led to the rapid retreat of glaciers, alterations in precipitation, and an increased frequency of extreme weather events such as floods and droughts [2]. This vulnerability highlights the pressing need to monitor and predict temperature trends to implement effective adaptation and mitigation strategies.

Understanding regional temperature anomalies requires access to high-quality, long-term climate data. To this end, Earth observation platforms, such as NASA's Prediction of Worldwide Energy Resources (POWER), provide an invaluable dataset encompassing surface meteorology and solar energy parameters [3]. Therefore, this research proposes a data-driven approach leveraging machine learning techniques on NASA POWER climate datasets to predict temperature anomalies and analyze the impacts of global warming in Nepal. By forecasting future temperature trends, this study aims to assist policymakers, environmentalists, and researchers in proactive decision-making and early warning systems.

II. Problem Statements

Nepal's diverse topography, ranging from tropical lowlands to the highest peaks of the Himalayas, makes it a complex region for climate modeling [4]. The continuous rise in average temperatures poses a severe threat to the cryosphere, hydrological cycles, and agricultural productivity. While global climate models provide macroscopic predictions, they often lack the spatial resolution required to accurately predict localized temperature anomalies in topographically complex regions like Nepal.

Traditional meteorological monitoring heavily relies on ground-based weather stations. However, in Nepal, these stations are sparsely distributed and often absent in high-altitude and remote regions. This scarcity of continuous, long-term observational data hinders the comprehensive assessment of spatiotemporal temperature trends. Furthermore, conventional statistical methods struggle to capture the complex, non-linear relationships inherent in climate data. Thus, there is a critical need for advanced computational models capable of handling extensive satellite-derived meteorological datasets to uncover hidden patterns and accurately forecast future climate scenarios [5].

III. Objectives

The specific objectives of this research are as follows:

1) To extract, preprocess, and integrate long-term surface meteorology and climatology datasets (e.g., maximum/minimum temperature, precipitation, humidity, solar radiation) for Nepal from the NASA POWER data access viewer.
2) To conduct exploratory data analysis (EDA) to identify spatiotemporal correlations, seasonal variations, and historical trends of temperature anomalies indicating global warming.
3) To implement, train, and compare the performance of various machine learning algorithms (such as Random Forest, Long Short-Term Memory networks, and XGBoost) in predicting future temperature trends.
4) To utilize the best-performing model to forecast temperature scenarios up to the year 2050, evaluating the potential environmental implications for the region.

IV. Dataset Sources and Collections

A. NASA POWER Data Access Viewer

This research utilizes the NASA POWER (Prediction of Worldwide Energy Resources) dataset, which is derived from satellite observations and assimilation models [3]. The dataset

provides continuous, long-term surface meteorology and solar energy data globally. For this study, the dataset is spatially filtered to cover the geographic boundary of Nepal.

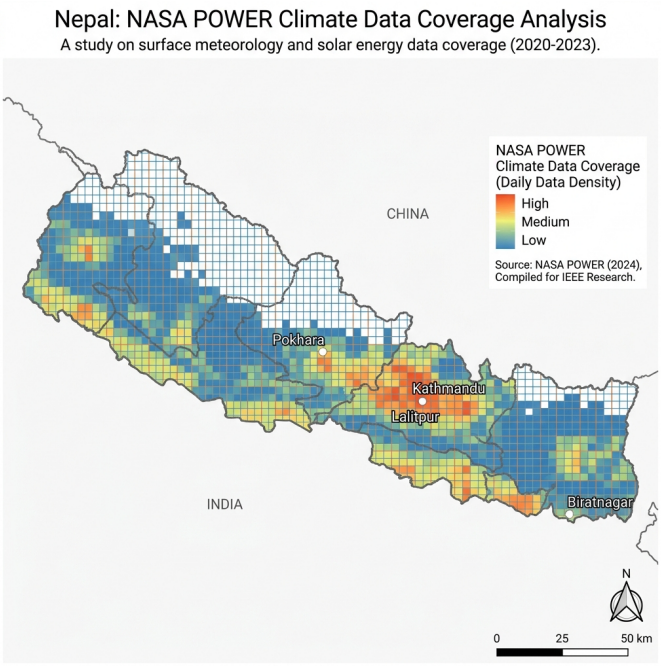


Fig. 1. NASA POWER Climate Data Coverage for Nepal

The acquired data includes the following crucial parameters:

- Geographic location (latitude, longitude, elevation)
- Temperature (Daily Max, Min, and Mean at 2m)
- Precipitation and Relative Humidity
- Earth Skin Temperature and Soil Temperature
- Solar Irradiance and Wind Speed

Source link: https://power.larc.nasa.gov/data-access-viewer/

V. Proposed Methodology

The proposed methodology encompasses a systematic data science pipeline, beginning with the acquisition of NASA POWER data through API integration, followed by rigorous data preprocessing to handle missing values and scale features. Subsequent phases involve feature engineering to extract temporal patterns, followed by the deployment and evaluation of robust machine learning models to forecast future temperature trajectories.

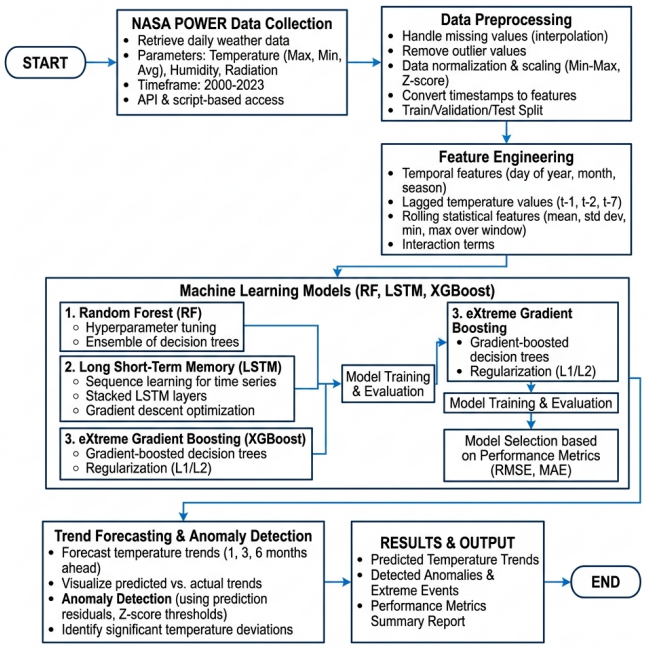


Fig. 2. Proposed Machine Learning Methodology Phases

VI. Expected Outcomes

This study will yield a comprehensive, preprocessed dataset and a robust framework tailored for analyzing Nepal's historical climate data derived from the NASA POWER platform. The exploratory analysis will map spatiotemporal hotspots of significant temperature increments and uncover key meteorological features driving these changes. By evaluating state-of-the-art machine learning models using standard metrics (such as MAE, RMSE, and R^2), the research will establish a highly accurate predictive system for regional climate variables.

Ultimately, the forecasted temperature anomalies up to the year 2050 will provide actionable insights into the trajectory of global warming impacts in Nepal. These findings will serve as a vital resource for environmental monitoring, helping policymakers and stakeholders formulate strategic interventions to mitigate climate risks and preserve the region's delicate ecological balance.

References

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